

Wieferich Primes and Hall's Conjecture

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Abstract

Nous montrons que la Conjecture de Hall pour courbes elliptiques implique l'existence d'une infinité de nombres premiers p satisfaisant $a^{p-1} \not\equiv 1 \pmod{p^{16}}$, pour chaque nombre entier a .

In the long and sinuous history of Fermat's last theorem, the following criterion stands out:

If a prime p satisfies

$$2^{p-1} \not\equiv 1 \pmod{p^2},$$

then the first case of Fermat's last theorem is true for the exponent p .

This criterion, proven by Wieferich [1] in 1909, was subsequently generalized by a number of authors to allow 2 to be replaced by any prime $q \leq 89$. (See [2].) Computational evidence shows that the only primes $p \leq 4 \times 10^{12}$ satisfying $2^{p-1} \equiv 1 \pmod{p^2}$ are $p = 1093$ and 3511 . Furthermore, for any integer a , the primes satisfying $a^{p-1} \equiv 1 \pmod{p^2}$ are very few. (Tables listing all solutions of $a^{p-1} \equiv 1 \pmod{p^2}$ for $2 \leq a \leq 99$, $3 \leq p \leq 2^{32}$ are listed in [3], p.347-349. The smallest known example where this congruence is satisfied is the case of $3^{10} \equiv 1 \pmod{11^2}$.) However, the following conjecture remains outstanding:

Conjecture 1. For any $a \in \mathbb{Z}$, there exist infinitely many primes p such that

$$a^{p-1} \not\equiv 1 \pmod{p^2}.$$

It is not known whether the assertion holds for any fixed integer a . Neither is the truth of the following weaker assertion.

Conjecture 2. For a given $a \in \mathbb{Z}$, there exists a natural number n such that there exist infinitely many primes p such that

$$a^{p-1} \not\equiv 1 \pmod{p^n}.$$

Silverman [4] has shown that the *ABC*-conjecture of Masser and Oesterlé implies Conjecture 1. More precisely, he showed that the *ABC*-conjecture implies that

$$\#\{p \leq x | a^{p-1} \not\equiv 1 \pmod{p^2}\} \gg_a \log x \text{ as } x \rightarrow \infty.$$

Another consequence of the *ABC*-conjecture is the following conjecture of Hall [5]:

Conjecture 3. (Hall): Let $\epsilon > 0$ be given. Then for any nonzero integer D , every solution $x, y \in \mathbb{Z}$ to the equation

$$y^2 = x^3 + D$$

satisfies

$$|x| \ll_{\epsilon} |D|^{2+\epsilon}.$$

We will show in this note that Conjecture 3 (which is *much* weaker than the powerful *ABC*-conjecture) implies Conjecture 2. In fact, we will prove the following:

Theorem. *If Hall's Conjecture (Conjecture 3) is true, then*

$$\#\{p \leq x | a^{p-1} \not\equiv 1 \pmod{p^{16}}\} \gg_a \frac{\log x}{\log \log x}.$$

We require two elementary lemmas.

Definition. Let $n \in \mathbb{Z}$. We will call n d^{th} -powerful if, for every $p|n$, we have $p^d|n$.

Lemma 1. *Let $n \in \mathbb{Z}$ be d^{th} -powerful. Then n may be written in the form*

$$n = \begin{cases} \alpha^2 \beta^d & \text{if } d \text{ is odd} \\ \alpha^2 \beta^{d+1} & \text{if } d \text{ is even} \end{cases},$$

for some integers α, β , with β squarefree.

Proof. Let $n = \prod_{i \in S} p_i^{e_i}$ be the canonical factorization of n . By hypothesis, each $e_i \geq d$.

If d is odd, we may write

$$\begin{aligned} n &= \prod_{e_i \text{ even}} p_i^{e_i} \prod_{e_j \text{ odd}} p_j^{e_j} = \prod_{e_i \text{ even}} p_i^{e_i} \prod_{e_j \text{ odd}} p_j^{e_j-d} \prod_{e_j \text{ odd}} p_j^d \\ &= \left(\prod_{e_i \text{ even}} p_i^{\frac{e_i}{2}} \prod_{e_j \text{ odd}} p_j^{\frac{e_j-d}{2}} \right)^2 \left(\prod_{e_j \text{ odd}} p_j \right)^d \end{aligned}$$

If d is even, write

$$n = \prod_{e_i=d} p_i^{e_i} \cdot m,$$

where $m = \prod_{e_i>d} p_i^{e_i}$ is $(d+1)^{th}$ -powerful. By the above, $m = \alpha^2 \beta^{d+1}$, for some $\alpha, \beta \in \mathbb{Z}$, β squarefree. Therefore,

$$n = \left(\prod_{e_i=d} p_i^{d/2} \alpha \right)^2 \beta^{d+1}.$$

Lemma 2. *Suppose that $p^t \parallel a^n - 1$ with $t \geq 1$. (i.e., p^t is the highest power of p dividing $a^n - 1$.) Then $a^{p-1} \not\equiv 1 \pmod{p^{t+1}}$.*

Proof. Let d denote the multiplicative order of $a \pmod{p}$. Then $a^n \equiv 1 \pmod{p} \Rightarrow d|n \Rightarrow a^d - 1 | a^n - 1$ so that $a^d \not\equiv 1 \pmod{p^{t+1}}$.

Let $p^s \parallel a^d - 1$ (so that $1 \leq s \leq t$) and write $a^d = 1 + kp^s$, where $p \nmid k$. Now, $d|p-1$, say $p-1 = df$. Thus,

$$\begin{aligned} a^{p-1} &= (a^d)^f = (1 + kp^s)^f \\ &= \sum_{i=0}^f \binom{f}{i} (kp^s)^i \\ &\equiv 1 + fkp^s \pmod{p^{s+1}} \\ &\not\equiv 1 \pmod{p^{t+1}} \end{aligned}$$

Proof (of Theorem). We show that, for any $a \in \mathbb{Z}$, $a^{3k+1} - 1$ is not 16^{th} -powerful, for all k sufficiently large. By the previous lemma, this will establish that, for all $p|a^{3k+1} - 1$, $a^{p-1} \not\equiv 1 \pmod{p^{16}}$.

Suppose that $a^{3k+1} - 1 = \alpha_k^2 \beta_k^d$, where d is odd. (By Lemma 1, any integer that is at least $(d-1)^{th}$ -powerful may be written in this form.) Then, multiplying through by $a^2 \beta_k^3$, we obtain

$$a^{3k+3} \beta_k^3 - a^2 \beta_k^3 = a^2 \alpha_k^2 \beta_k^{d+3}$$

or

$$(a^{k+1} \beta_k)^3 - a^2 \beta_k^3 = (a \alpha_k \beta_k^{\frac{d+3}{2}})^2.$$

Applying Hall's Conjecture, we have

$$a^{k+1} \beta_k \ll_{\epsilon} (a^2 \beta_k^3)^{2+\epsilon} \text{ or } a^{k+1} \ll_{\epsilon, a} \beta_k^{5+\epsilon}.$$

Then

$$\beta_k^d < \alpha_k^2 \beta_k^d < a^{3k+1} - 1 < (a^{k+1})^3 \ll_{\epsilon, a} \beta_k^{15+3\epsilon}.$$

Now, for fixed β , the curve $\beta^d y^2 = ax^3 - 1$ has but finitely many integral points. Therefore, $\beta_k \rightarrow \infty$ as $k \rightarrow \infty$. We can thus conclude from the above inequality that $d \leq 15$, for all sufficiently large k . Thus, for all sufficiently large k , $a^{3k+1} - 1$ is not 16^{th} -powerful $\Rightarrow$ for all $p | a^{3k+1} - 1$, $a^{p-1} \not\equiv 1 \pmod{p^{16}}$. Since for $\gcd(3k+1, 3l+1) = 1$, $\gcd(a^{3k+1} - 1, a^{3l+1} - 1) = a - 1$, every sufficiently large prime $q \equiv 1 \pmod{3}$ yields a distinct prime $p | \frac{a^q - 1}{a - 1}$. There are thus, infinitely many primes p such that $a^{p-1} \not\equiv 1 \pmod{p^{16}}$. In fact,

$$\begin{aligned} \#\{p \leq x | a^{p-1} \not\equiv 1 \pmod{p^{16}}\} &\gg \#\{p \equiv 1 \pmod{3} | a^p - 1 \leq x\} \\ &\gg \#\{p \equiv 1 \pmod{3} | p \leq \frac{\log x}{\log a}\} \\ &\gg_a \frac{\log x}{\log \log x} \text{ as } x \rightarrow \infty, \end{aligned}$$

by the prime number theorem for arithmetic progressions.

References

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